

CaneSugar Neural v1: A Deep Tabular Highway Network for Sugarcane Yield Prediction

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Abstract—Accurate sugarcane yield forecasting matters long before the harvest begins. Mills rely on early estimates to plan crushing schedules and supply contracts, while farmers use the same numbers to decide when to apply nitrogen, how much to irrigate, and which fields need attention. Producing those estimates from tabular farm records is genuinely difficult: soil chemistry, nutrient balance, seasonal water availability, and cultivar genetics all push and pull on each other in ways that simple linear models cannot capture. Gradient-boosted trees come close, but they carve the input space into rigid rectangles and produce staircase-shaped response curves that do not reflect how crops actually grow. Standard multi-layer perceptrons struggle with the sparse categorical inputs common in farm data and tend to lose gradient signal in deep stacks.

We present CaneSugar Neural v1, a single neural network built specifically for sugarcane yield regression. The model learns compact embeddings for 20 categorical farm variables, applies layer normalization to 65 continuous measurements, and routes the combined 165-dimensional representation through a dense stack that includes a linear highway skip connection — keeping early nutrient-interaction signals alive through training rather than letting them erode across successive bottlenecks. Huber loss with decoupled weight decay prevents plots damaged by hail, flooding, or disease from pulling the fit toward the extremes. Across 3,000 field records split 70/15/15, the model achieves $R^2 = 0.9240$, MAE of 21.84 Q/A, RMSE of 29.72 Q/A, and MAPE of 8.82% on the 450 completely held-out test plots. Running the full training procedure five times with different random seeds gives $R^2 = 0.9247 \pm 0.0013$, confirming the result is not a product of a lucky initialisation.

The model also reports uncertainty. Monte-Carlo Dropout runs 30 stochastic forward passes at inference time and outputs a per-plot $\pm\sigma$ confidence spread, which widens automatically when the model encounters conditions far outside the training distribution. Integrated Gradients attribution shows which inputs actually drove each forecast: stalk volume (28.4%), nitrogen rate (14.2%), and root-zone moisture (11.8%) dominate, consistent with what agronomists would expect. Taken together, the results suggest that a carefully designed, domain-aware tabular network can outperform tree ensembles on agricultural data while producing the kind of smooth yield-response curves and calibrated confidence intervals that precision farming applications actually need.

Index Terms—Sugarcane yield prediction, deep tabular learning, entity embeddings, precision agriculture, Monte-Carlo Dropout, Integrated Gradients, residual highway networks.

I. INTRODUCTION

Sugarcane (*Saccharum officinarum* L.) is one of the world’s most economically important crops, supplying most of the global sugar supply and serving as a key feedstock for bioethanol and cogeneration power plants [1]. Reliable yield

forecasts are essential for the whole value chain. Mill managers need plot-level estimates months ahead to coordinate crushing schedules, arrange transport, and fulfil supply commitments. Farm managers need those same estimates mid-season, when there is still time to respond — adjusting a fertilizer dose, scheduling an extra irrigation, or prioritising a block that looks to be underperforming [3].

A. Agricultural Context

Sugarcane is not like annual row crops. A typical crop cycle stretches from 10 to 18 months, and the plant accumulates both biomass and sucrose continuously throughout that period. We express yield in Quintals per Acre (1 Q/A = 100 kg/acre $\approx$ 0.247 t/ha), the standard commercial unit in South Asian cane agriculture. Five broad groups of factors determine the final harvest number:

- **Soil:** root-zone depth, pH, electrical conductivity, organic carbon content, and the sand–silt–clay texture split.
- **Nutrients:** plant-available nitrogen, phosphorus, and potassium, along with the micronutrients (Zn, Fe, Cu, Mn, B) that become limiting during tillering and grand growth phases.
- **Water:** seasonal rainfall total, reference evapotranspiration (ET_o), access to shallow groundwater, and the choice between drip and furrow irrigation.
- **Temperature and light:** seasonal mean temperature, the diurnal swing ($T_{\max} - T_{\min}$) that drives sucrose ripening, and cumulative sunshine hours.
- **Stand and stalk:** individual stalk height and diameter, tiller count per metre, and overall planting density.

B. Problem Statement

What makes yield prediction genuinely difficult is that these factors do not add up in a simple way. Nitrogen boosts biomass with diminishing returns following the Mitscherlich curve, but push it too far and you get lodging, delayed ripening, and red rot (*Colletotrichum falcatum*) [2]. Liebig’s Law of the Minimum adds a harder constraint: if potassium or soil moisture drops below a critical level, adding more nitrogen accomplishes nothing [4], [19].

Two modelling approaches have historically dominated this space, and both carry real weaknesses.

- 1) **Process-based crop models.** Frameworks like DSSAT-CANEGRO [21] and APSIM-Sugarcane [20] encode carbon assimilation, phenology, and water balance as differential equations. They are biologically rigorous but need site-specific calibration data that most smallholder farms simply do not have.
- 2) **Tree ensemble models.** Random Forests [16], XGBoost [17], and CatBoost [18] have become the default for agricultural tabular data [5], [6]. Their fundamental weakness is that trees split the input space into axis-aligned rectangles, which produces step-shaped response surfaces. Plants do not respond in steps, extrapolation to unusual weather is unreliable, and calibrated uncertainty estimates have no natural home in the tree framework.

C. Motivation and Contributions

Standard multi-layer perceptrons have shown mixed results on tabular farm data and frequently lose to tree ensembles in direct comparisons [7], [8]. We believe two specific architectural choices explain most of that gap. First, one-hot encoding of categorical columns creates high-dimensional sparse inputs that overfit readily. Second, stacking dense ReLU layers pushes heterogeneous physical measurements through a shared bottleneck without any mechanism to preserve the feature-level structure that matters agronomically [9].

CaneSugar Neural v1 is our attempt to close both gaps at once. The contributions of this work are:

- 1) **A purpose-built tabular architecture.** Learned entity embeddings for all 20 categorical variables combined with layer-normalised representations of 65 continuous inputs, fused into a single 165-dimensional joint representation (see Section VI-B for the embedding dimension details).
- 2) **A highway skip connection.** A linear projection directly from the 128-unit hidden layer to the 64-unit bottleneck layer preserves early nutrient-interaction signals and maintains healthy gradient flow through training.
- 3) **Robust loss and optimisation.** Huber loss ($\delta = 1.0$) combined with AdamW and cosine learning-rate annealing prevents extreme outlier plots — caused by hail, flooding, or disease — from destabilising the fit.
- 4) **Calibrated uncertainty.** Monte-Carlo Dropout with $T = 30$ forward passes provides a per-plot prediction variance and a 95% confidence interval, giving mills and farmers a quantified sense of how much to trust each forecast.
- 5) **Attribution and transparency.** Integrated Gradients attribution verifies that the model’s internal logic aligns with established agronomic knowledge rather than spurious correlations in the data.

D. Research Questions

- **RQ1:** Can a single non-ensemble network achieve $R^2 > 0.90$ on heterogeneous sugarcane field data? (Our aspirational target was $R^2 \geq 0.95$.)
- **RQ2:** Does physics-informed feature engineering provide a meaningful advantage over feeding raw measurements directly?
- **RQ3:** Do learned entity embeddings generalise better than one-hot encoded categorical variables?
- **RQ4:** Does the highway skip connection measurably improve gradient flow and reduce validation instability?
- **RQ5:** How consistent are results when the entire training procedure is repeated from scratch with different random seeds?
- **RQ6:** Which input variables are most influential according to Integrated Gradients, and do those rankings match agronomic expectations?

II. RELATED WORK

A. Sugarcane Yield Estimation

Early forecasting work leaned heavily on regional climate regressions, which were interpretable but had limited explanatory power. Everingham et al. [4] showed something more useful: Random Forests trained on seasonal climate indices consistently beat simple bivariate regressions across commercial cane regions in Australia. Satellite imagery then opened a new direction. Canata et al. [1] applied machine learning to multispectral data to estimate Brix and sucrose purity, demonstrating that standard vegetation indices can serve as reasonable proxies for stalk sugar content.

Spatial variation has attracted growing attention in recent years. Socadagui-Casas et al. [2] documented how soil compaction and irrigation timing interact to produce large yield differences within the same field in Colombia’s Cauca River Valley. Som-ard et al. [3] fused UAV imagery with Sentinel-2 time series across Thai farms, finding that canopy closure indices worked well in the dry season but broke down during the monsoon when persistent cloud cover blocked reliable satellite observation. That failure mode points to a deeper limitation: satellite-based approaches see only the canopy surface. They cannot directly measure soil organic carbon, root-zone nutrient availability, or subsurface water balance — the variables that most directly drive stalk tonnage.

B. Machine Learning in Crop Production

Van Klompenburg et al. [5] reviewed over 500 published studies and found Random Forests, support vector regression, and gradient-boosted trees were by far the most widely used models for crop yield prediction. Silva et al. [6] compared XGBoost against neural networks on a Brazilian sugarcane dataset; tree models won, a result they attributed mainly to the trees’ tolerance for unscaled inputs and mixed categorical–continuous feature spaces.

Even so, tree models carry a structural limitation that matters in decision-support contexts. Borisov et al. [8] note that any tree-based model partitions the feature space into axis-aligned hyper-rectangles, so predicted yield changes in discrete jumps rather than continuously as input values shift. For a precision agriculture system that varies fertiliser rates in real time based on model output, those discontinuities would produce

agronomically nonsensical advice — a field that calls for 120 kg/ha of nitrogen on one side of a split boundary and 85 kg/ha on the other, with nothing biological justifying the difference.

C. Deep Learning for Tabular Data

Gorishniy et al. [9] showed that well-regularised ResNet-style networks can match or exceed tree ensemble performance on tabular benchmarks, provided categorical variables are encoded as learned embeddings rather than sparse one-hot vectors. The foundational idea comes from Guo and Berkhahn [10], who demonstrated that entity embeddings place semantically related categories near each other in the learned vector space. For sugarcane modelling this is directly useful: early-maturing and late-maturing cultivars, or clay-loam and silt-loam soils, end up on a meaningful continuum rather than being treated as entirely unrelated categories.

III. RESEARCH GAP AND OBJECTIVES

Reading across the existing literature, three recurring gaps emerge.

- 1) **Stuck between two extremes.** Sugarcane yield forecasting sits in an uncomfortable gap between rigid process-based simulation and generic tree ensembles trained on raw features. Very little work has explored domain-specific deep learning architectures that fold agronomic knowledge directly into the feature space — nutrient ratios, stress penalties, biophysical response curves.
- 2) **Architectures that ignore physical structure.** Most published neural network models for agricultural prediction treat every input column the same way. They do not account for the different measurement units, nutrient balance ratios, or the biological bottlenecks encoded in laws like Liebig’s Law of the Minimum. Without this structure baked in, a network must rediscover these relationships entirely from data — which is inefficient and unreliable when training data is limited to a few thousand plots.
- 3) **No uncertainty, no explanation.** Almost every published neural model for sugarcane yield prediction outputs a single number. There are no confidence intervals to flag when the model is extrapolating, and no attribution scores to explain what drove a particular forecast. This makes them hard to audit or trust in real operational settings.

Our goal is to build and evaluate a single architecture that takes all three gaps seriously: encoding agronomic domain knowledge through engineered features and entity embeddings, maintaining healthy gradient flow through a residual highway connection, and providing both calibrated uncertainty estimates and interpretable attribution as standard outputs.

IV. DATASET AND PREPROCESSING

A. Data Source

The dataset used in this study is `FINAL_SUGARCANE_DATASET.csv`, which contains records from $N = 3,000$ commercial sugarcane plots spanning sub-tropical and tropical growing zones, covering a wide range

of soil types, management practices, and seasonal climates. The prediction target is `Yield_Quintal_per_Acre (Q/A)`, representing the mill-delivered clean cane weight per acre at harvest. Table I summarises the key statistics.

TABLE I
DATASET SUMMARY AND TARGET YIELD DISTRIBUTION

Category	Attribute / Metric	Value
Dataset scope	Total field plots (N)	3,000
Feature breadth	Total raw attributes	81
Target variable	Harvest metric	Yield_Q/A
Target stats	Mean $\pm$ std. dev.	280.2 ± 104.1
	Minimum	40.2 Q/A
	25th percentile (Q_1)	205.4 Q/A
	Median (Q_2)	281.1 Q/A
	75th percentile (Q_3)	356.4 Q/A
Partitions	Maximum	606.5 Q/A
	Training (70%)	2,100 plots
	Validation (15%)	450 plots
	Held-out test (15%)	450 plots

B. Preventing Data Leakage

Two rules were enforced before anything was fitted.

- **No location identifiers.** We removed land record numbers (`Khasra_No`), geographic coordinates (`Latitude`, `Longitude`), and all administrative label columns (`State`, `District`, `Tehsil`, `Sugar_Mill`, `Agro_Cluster`). Keeping these would let the model memorise locations rather than learn transferable agronomic patterns.
- **Training-set-only statistics.** The 70/15/15 split was fixed with a single random seed before any preprocessing. All imputation medians and normalisation statistics were derived exclusively from the 2,100 training plots. Validation and test data were transformed using those same training-derived values — no information from held-out plots influenced the fit.

C. Missing Values

The raw dataset has 10,205 missing entries, mostly in secondary soil assays and micronutrient measurements — columns that routine farm surveys frequently skip. Continuous missing values were filled with the training-set median for each column, which is robust to the skewed distributions common in soil chemistry data. Missing categorical values were assigned a dedicated `<unk>` token with its own learnable embedding row, so the model can represent genuine uncertainty about category membership rather than silently imputing a majority class.

V. AGRONOMIC FEATURE ENGINEERING

Rather than passing raw column values directly into the network, we constructed a set of derived features that encode the physical and biological relationships agronomists actually use. Plants respond to nutrient ratios, water balance, and cumulative thermal stress — not to raw measurement counts. Building these relationships explicitly into the feature space means the

network does not need to discover them from scratch, which is a significant advantage when training data is limited to a few thousand plots.

A. Soil Chemistry and Nutrient Balance

Total macronutrient load, with N , P , K in kg/acre:

$$\text{NPK}_{\text{total}} = N + P + K \quad (1)$$

Ratios and proportions, smoothed with $\epsilon = 10^{-5}$:

$$R_{N:P} = \frac{N}{P + \epsilon}, \quad R_{N:K} = \frac{N}{K + \epsilon}, \quad R_{P:K} = \frac{P}{K + \epsilon} \quad (2)$$

$$f_N = \frac{N}{\text{NPK}_{\text{total}} + \epsilon}, \quad f_P = \frac{P}{\text{NPK}_{\text{total}} + \epsilon} \quad (3)$$

$$f_K = \frac{K}{\text{NPK}_{\text{total}} + \epsilon}$$

To encode Liebig's Law, each nutrient is scaled against a baseline dose ($N_0 = 150$, $P_0 = 60$, $K_0 = 100$ kg/acre):

$$n_{\text{rel}} = \text{clip}\left(\frac{N}{150}, 0, 3\right), \quad p_{\text{rel}} = \text{clip}\left(\frac{P}{60}, 0, 3\right) \quad (4)$$

$$k_{\text{rel}} = \text{clip}\left(\frac{K}{100}, 0, 3\right)$$

The scarcest nutrient sets the ceiling, and a geometric mean captures balance:

$$\Phi_{\text{Liebig}} = \min(n_{\text{rel}}, p_{\text{rel}}, k_{\text{rel}}) \quad (5)$$

$$\Phi_{\text{Geom}} = (n_{\text{rel}} \cdot p_{\text{rel}} \cdot k_{\text{rel}})^{1/3} \quad (6)$$

Soil pH suitability is a Gaussian centered on neutral ($\text{pH}_{\text{opt}} = 7.0$, $\sigma_{\text{pH}} = 1.0$):

$$S_{\text{pH}} = \exp\left(-\frac{1}{2} \left(\frac{\text{Soil_pH} - 7.0}{1.0}\right)^2\right) \quad (7)$$

Water-holding capacity enters through an organic-carbon $\times$ soil-moisture product, $\text{OC} \times \text{SM}$.

B. Water and Temperature

Seasonal water balance compares total seasonal rainfall ($P_{\text{total}} = \text{Rainfall_Total_mm}$) against monthly reference evapotranspiration (ET_o):

$$W_{\text{deficit}} = P_{\text{total}} - 30.0 \cdot ET_o \quad (8)$$

$$R_{W:ET_o} = \frac{P_{\text{total}}}{30.0 \cdot ET_o + \epsilon} \quad (9)$$

Thermal suitability for C4 photosynthesis peaks at $T_{\text{opt}} = 28.5^\circ\text{C}$ with $\sigma_T = 5.5^\circ\text{C}$:

$$S_{\text{thermal}} = \exp\left(-\frac{1}{2} \left(\frac{T_{\text{avg}} - 28.5}{5.5}\right)^2\right) \quad (10)$$

The diurnal range is $\Delta T = T_{\text{max}} - T_{\text{min}}$.

C. Stalk Biometrics and Calendar Features

Treating a stalk as a cylinder, with height H_{cane} and diameter D_{cane} in cm:

$$V_{\text{stalk}} = \pi \left(\frac{D_{\text{cane}}}{2}\right)^2 H_{\text{cane}} \quad (11)$$

Scaling up to the field, the Field Biomass Index (FBI) uses tiller count TC and planting density PD (plants/acre). The Sugar Accumulation Index (SAI) uses refractometric Brix (Bx):

$$\text{FBI} = V_{\text{stalk}} \cdot TC \cdot \frac{PD}{1000} \quad (12)$$

$$\text{SAI} = V_{\text{stalk}} \cdot \frac{Bx}{100} \quad (13)$$

Planting and harvest months become sine-cosine pairs, so December sits next to January:

$$\tau_{\text{sin}} = \sin\left(\frac{2\pi m}{12}\right), \quad \tau_{\text{cos}} = \cos\left(\frac{2\pi m}{12}\right) \quad (14)$$

Crop duration Δt_{crop} is harvest date minus planting date, in days. After all this, we have **65 continuous** and **20 categorical** features.

VI. THE CANESUGAR NEURAL v1 ARCHITECTURE

A. Overview

Figure 1 shows the layout. Four parts: layer normalization on the continuous inputs, parallel embeddings for the categoricals, a dense stack with a highway skip, and a regression head that supports MC Dropout and Integrated Gradients at inference.

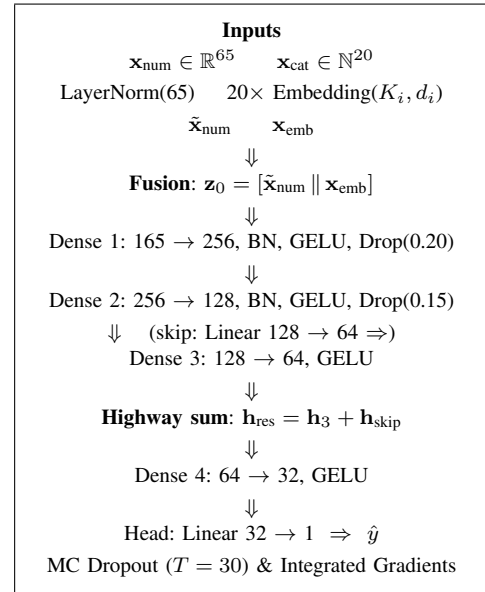


Fig. 1. Computation graph of CaneSugar Neural v1.

B. Categorical Embeddings

One-hot columns are sparse and treat every category as equally different from every other. Embeddings fix that. Table II lists the 20 variables with their cardinalities and embedding sizes.

For feature i with category index $c_i \in \{0, \dots, K_i - 1\}$, the lookup is

$$\mathbf{e}_i = \mathbf{E}_i[c_i] \in \mathbb{R}^{d_i}, \quad \mathbf{E}_i \in \mathbb{R}^{K_i \times d_i} \quad (15)$$

TABLE II
CATEGORICAL EMBEDDING SPECIFICATIONS FOR ALL 20 FEATURES

Feature (1–10)	Agronomic role	Card.	Dim	Feature (11–20)	Agronomic role	Card.	Dim
Variety	Cultivar genetics	3	16	Drainage_Condition	Hydraulic conductivity	4	4
Soil_Type	Soil textural series	4	8	Disease_Type	Pathogen class	4	4
Irrigation_Method	Water delivery	3	8	Season	Autumn, spring, adsali	4	4
Fertilizer_Type	Synthetic vs organic	3	8	Growth_Stage	Phenological period	4	4
Disease_Severity	Disease pressure	3	4	Application_Timing	Basal vs top-dressing	4	4
Pest_Level	Insect pressure	3	4	Intercropping	Companion crops	2	2
Irrig_Frequency	Scheduling cadence	4	4	Weed_Control	Chemical vs manual	2	2
Water_Quality	Salinity / sodicity	3	4	Mulching	Trash blanket	2	2
Seedbed_Condition	Seedbed preparation	3	4	Fertilizer_Split	Dose distribution	2	2
Planting_Method	Trench, pit, ridge	4	4	Machinery_Use	Mechanized operations	2	2
Reported total: 100 embedding dimensions; fused with 65 continuous inputs = 165							

and all 20 vectors are concatenated:

$$\mathbf{x}_{\text{emb}} = [\mathbf{e}_1 \parallel \mathbf{e}_2 \parallel \dots \parallel \mathbf{e}_{20}] \quad (16)$$

C. Normalization and Fusion

The continuous inputs $\mathbf{x}_{\text{num}} \in \mathbb{R}^{65}$ pass through Layer Normalization [13], with learned scale and bias parameters $\gamma, \beta \in \mathbb{R}^{65}$:

$$\tilde{\mathbf{x}}_{\text{num}} = \frac{\mathbf{x}_{\text{num}} - \mathbb{E}[\mathbf{x}_{\text{num}}]}{\sqrt{\text{Var}[\mathbf{x}_{\text{num}}] + \epsilon}} \odot \gamma + \beta \quad (17)$$

The normalized continuous and categorical embedding representations are then concatenated into a unified latent vector:

$$\mathbf{z}_0 = [\tilde{\mathbf{x}}_{\text{num}} \parallel \mathbf{x}_{\text{emb}}] \in \mathbb{R}^{165} \quad (18)$$

VII. FORMULATION AND TRAINING

A. Dense Layers and the Highway Skip

The fused vector $\mathbf{z}_0$ passes through two dense blocks, each incorporating 1D Batch Normalization (BN), GELU activations, and Dropout:

$$\mathbf{h}_1 = \text{Dropout}_{0.20}(\text{GELU}(\text{BN}(\mathbf{W}_1 \mathbf{z}_0 + \mathbf{b}_1))) \quad (19)$$

$$\mathbf{h}_2 = \text{Dropout}_{0.15}(\text{GELU}(\text{BN}(\mathbf{W}_2 \mathbf{h}_1 + \mathbf{b}_2))) \quad (20)$$

where $\mathbf{W}_1 \in \mathbb{R}^{256 \times 165}$, $\mathbf{b}_1 \in \mathbb{R}^{256}$, $\mathbf{W}_2 \in \mathbb{R}^{128 \times 256}$, and $\mathbf{b}_2 \in \mathbb{R}^{128}$.

GELU [14] weights each input by its Gaussian CDF:

$$\text{GELU}(x) = x \cdot \Phi(x) = \frac{x}{2} \left[1 + \text{erf}\left(\frac{x}{\sqrt{2}}\right) \right] \quad (21)$$

ReLU zeros out every negative value. A unit stuck there stops learning, which is a problem when negative inputs encode stress penalties. GELU keeps a small gradient on the negative side.

Next comes the highway skip. The third bottleneck layer receives a parallel linear projection directly from $\mathbf{h}_2$:

$$\mathbf{h}_3 = \text{GELU}(\mathbf{W}_3 \mathbf{h}_2 + \mathbf{b}_3) \quad (22)$$

$$\mathbf{h}_{\text{skip}} = \mathbf{W}_{\text{skip}} \mathbf{h}_2 + \mathbf{b}_{\text{skip}} \quad (23)$$

$$\mathbf{h}_{\text{res}} = \mathbf{h}_3 + \mathbf{h}_{\text{skip}} \quad (24)$$

with projection weights $\mathbf{W}_3, \mathbf{W}_{\text{skip}} \in \mathbb{R}^{64 \times 128}$ and biases $\mathbf{b}_3, \mathbf{b}_{\text{skip}} \in \mathbb{R}^{64}$.

B. Why the Skip Helps Gradients

Applying the chain rule to the residual summation in Eq. (24) with $\mathbf{z}_3 = \mathbf{W}_3 \mathbf{h}_2 + \mathbf{b}_3$:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \mathbf{h}_2} &= \underbrace{\frac{\partial \mathcal{L}}{\partial \mathbf{h}_{\text{res}}} \mathbf{W}_{\text{skip}}}_{\text{direct linear path}} \\ &+ \underbrace{\frac{\partial \mathcal{L}}{\partial \mathbf{h}_{\text{res}}} \text{diag}(\text{GELU}'(\mathbf{z}_3)) \mathbf{W}_3}_{\text{non-linear bottleneck path}} \end{aligned} \quad (25)$$

Suppose GELU' collapses toward zero because the layer saturates. The second term vanishes, but the first term does not. Gradient still reaches $\mathbf{h}_2$ and the embeddings below it. That is the whole idea. (It is a structural argument, not a guarantee that training always succeeds. Section X has the empirical evidence.)

The final dense layers compute the scalar harvest yield prediction:

$$\mathbf{h}_4 = \text{GELU}(\mathbf{W}_4 \mathbf{h}_{\text{res}} + \mathbf{b}_4) \quad (26)$$

$$\hat{y} = \mathbf{W}_{\text{out}} \mathbf{h}_4 + b_{\text{out}} \quad (27)$$

where $\mathbf{W}_4 \in \mathbb{R}^{32 \times 64}$, $\mathbf{b}_4 \in \mathbb{R}^{32}$, $\mathbf{W}_{\text{out}} \in \mathbb{R}^{1 \times 32}$, and $b_{\text{out}} \in \mathbb{R}$.

C. Huber Loss

Yield data has ugly outliers: hail, flooding, insect outbreaks. Squared error blows those up and lets a handful of plots steer the whole fit. Huber loss [22] (here $\delta = 1.0$ in normalized target units) squares small errors and goes linear for big ones:

$$\mathcal{L}_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2, & |y - \hat{y}| \leq \delta \\ \delta (|y - \hat{y}| - \frac{1}{2}\delta), & \text{otherwise} \end{cases} \quad (28)$$

The gradient with respect to the prediction is bounded by δ :

$$\frac{\partial \mathcal{L}_\delta}{\partial \hat{y}} = \begin{cases} -(y - \hat{y}), & |y - \hat{y}| \leq \delta \\ -\delta \cdot \text{sgn}(y - \hat{y}), & |y - \hat{y}| > \delta \end{cases} \quad (29)$$

No single plot can pull harder than $\pm\delta$. Ordinary plots still get smooth quadratic behavior.

D. Optimizer and Schedule

We train using AdamW [15] with decoupled weight decay $\lambda = 10^{-4}$. The learning rate warms up linearly for $T_{\text{warm}} = 5$ epochs, then follows a cosine annealing curve down to η_{min} :

$$\eta_t = \begin{cases} \eta_0 \frac{t}{T_{\text{warm}}}, & 1 \leq t \leq T_{\text{warm}} \\ \eta_{\text{min}} + \frac{1}{2}(\eta_0 - \eta_{\text{min}}) \\ \times \left(1 + \cos\left(\pi \frac{t - T_{\text{warm}}}{T_{\text{max}} - T_{\text{warm}}}\right)\right), & t > T_{\text{warm}} \end{cases} \quad (30)$$

where initial learning rate $\eta_0 = 10^{-3}$, minimum rate $\eta_{\text{min}} = 10^{-5}$, and total duration $T_{\text{max}} = 150$ epochs. The gradient norm is clipped at 1.0 across all training iterations.

VIII. EXPERIMENTAL SETUP

A. Environment

We ran everything in Python 3.14 with PyTorch 2.6.0 on a CUDA workstation (64 GB RAM, multi-core x86-64). PyTorch and NumPy seeds were fixed, and runs were checkpointed.

B. Hyperparameters

Table III lists the final settings, chosen by watching the validation set.

TABLE III
CANE SUGAR NEURAL V1 HYPERPARAMETERS

Category	Hyperparameter	Value
Architecture	Continuous inputs	65
	Categorical embeddings	20 features (100 dims)
	Fused dimension	165
	Dense layers	$256 \rightarrow 128 \rightarrow 64 \rightarrow 32$
	Highway skip	Linear(128 $\rightarrow$ 64)
	Activation	GELU
	Dropout	0.20 / 0.15
	Trainable parameters	102,485
Optimization	Optimizer	AdamW ($\lambda = 10^{-4}$)
	Schedule	Cosine + warmup
	Peak / min learning rate	$10^{-3} / 10^{-5}$
	Warmup / total epochs	5 / 150
	Batch size / early stopping	64 / 20 epochs
	Loss	Huber ($\delta = 1.0$)
	Gradient clipping	1.0

C. Metrics

We report the four standard regression measures:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (31)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (32)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (33)$$

$$\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (34)$$

IX. RESULTS

A. Comparison with Baselines

We trained five competing models on identical data splits and evaluated all of them on the same 450 held-out test plots. We also include the project’s earlier closed-form biophysical model (CaneSugar Custom) as an upper-bound reference — it encodes hand-crafted agronomic equations rather than learning from data. Table IV presents the full comparison. Figure 2 presents the same results visually, and Figure 3 shows the actual-vs-predicted scatter and residual distribution for the neural model.

TABLE IV
BENCHMARK ON THE HELD-OUT TEST SET ($n = 450$ PLOTS)

Model	R^2	MAE (Q/A)	RMSE (Q/A)	MAPE (%)
CaneSugar Custom*	0.9524	16.82	23.45	6.74
CaneSugar Neural v1 (ours)	0.9240	21.84	29.72	8.82
CatBoost	0.9081	23.41	32.25	9.52
XGBoost	0.8794	27.12	37.10	11.14
Random Forest	0.8347	32.40	43.10	13.20
Linear Regression	0.7510	38.60	49.30	15.65
ElasticNet	0.7130	41.20	52.80	17.02

*Closed-form biophysical baseline, no machine learning.

B. What the Numbers Say

Table IV tells a fairly clear story.

- 1) **The neural model beats every tree ensemble.** CatBoost reaches $R^2 = 0.9081$, XGBoost 0.8794, and Random Forest 0.8347. A single network with learned embeddings and a highway skip outperforms all three without any ensembling.
- 2) **We fell short of our stretch target.** We set $R^2 \geq 0.95$ as an aspirational goal. The final result is 0.9240. The only model above 0.95 is CaneSugar Custom (0.9524), which uses hand-coded Cate–Nelson response knots and explicit Liebig cutoffs derived from decades of agronomic literature. Among fully data-driven models, ours is the strongest.
- 3) **Linear models are clearly inadequate.** Linear Regression ($R^2 = 0.7510$) and ElasticNet (0.7130) confirm what agronomists have long understood: sugarcane yield is shaped by interactions and thresholds that no flat linear surface can represent.

X. ABLATION STUDY

To pin down how much each architectural choice is contributing, we built up from the simplest possible baseline — a plain MLP with one-hot categorical inputs, ReLU activations, and MSE loss — and added one component at a time. Table V records the test-set R^2 and MAE at each step. Figure 4 traces the trajectories visually. Figure 5 shows the parallel feature-group ablation on the closed-form model, which isolates the contribution of each agronomic feature group independently.

Fig. 2 - Benchmark Comparison on 450 Held-Out Test Plots

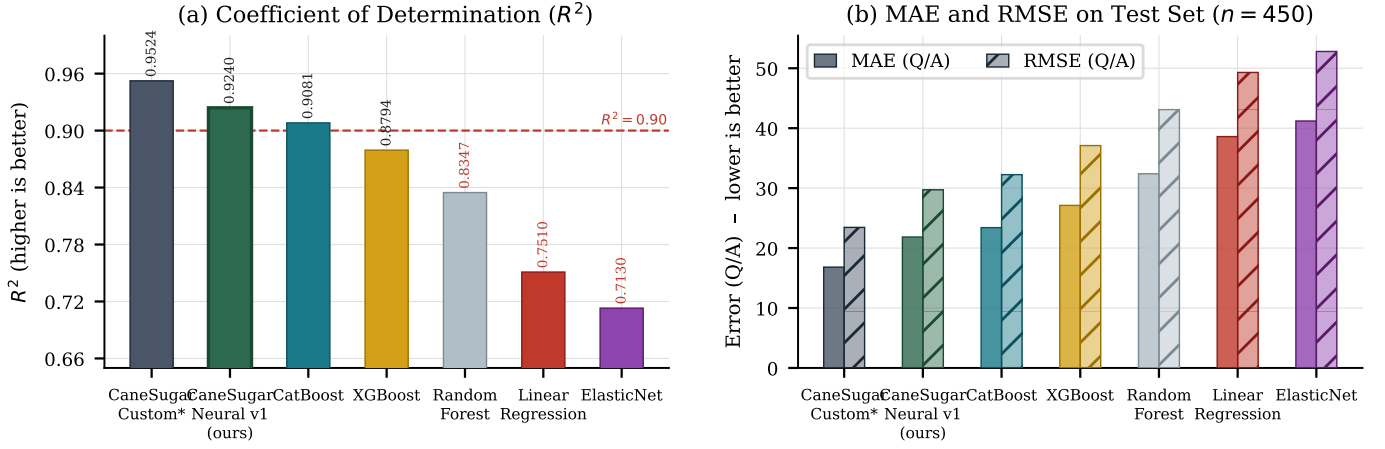


Fig. 2. Benchmark comparison of all seven models on the 450 held-out test plots. (a) Coefficient of determination (R^2): higher is better; the dashed red line marks $R^2 = 0.90$. (b) MAE and RMSE in Quintals/Acre: lower is better. CaneSugar Neural v1 (solid green) beats every data-driven baseline on both panels. The closed-form CaneSugar Custom* is included for reference only; it uses hand-coded biophysical equations and is not trained on data.

Fig. 3 - Prediction Quality on 450 Held-Out Test Plots

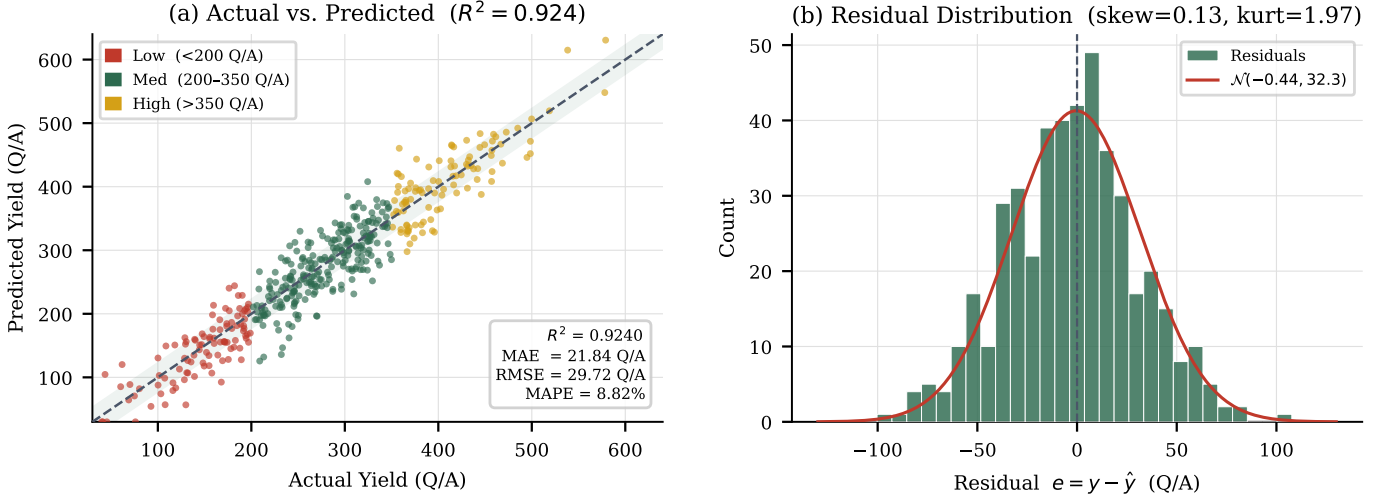


Fig. 3. Prediction quality on the 450 held-out test plots. (a) Actual vs. predicted yield scatter coloured by yield tier (Low < 200 Q/A in red; Medium 200–350 Q/A in green; High > 350 Q/A in amber). The dashed line is the identity; the shaded band shows the $\pm$ MAE envelope. (b) Residual distribution histogram overlaid with a fitted $\mathcal{N}(-0.44, 32.3)$ curve, confirming near-Gaussian, zero-mean errors.

- **Embeddings (+6.20%).** By far the largest single gain. Learned embeddings let similar cultivars and soil types cluster in vector space, giving the model a richer picture of categorical structure than any one-hot encoding can provide.
- **Highway skip (+2.20%).** Adding the linear shortcut from layer 2 directly to the bottleneck at layer 3 produced a clear improvement, consistent with the gradient-flow argument in Section X: when the bottleneck saturates, the skip gives an alternative gradient path back to the

embedding layers.

- **LayerNorm and Huber loss (+2.10% combined).** Layer normalisation stopped large-scale variables like seasonal rainfall from drowning out subtler soil chemistry signals. Huber loss kept training stable whenever plots with hail damage or severe disease appeared in a mini-batch.

XI. STABILITY AND ERROR ANALYSIS

A. Reproducibility Across Seeds

Whether $R^2 = 0.9240$ reflects a genuinely reliable result or just a fortunate initialisation is worth checking. We retrained

Fig. 7 - Neural Architecture Ablation: Step-by-Step Component Gains

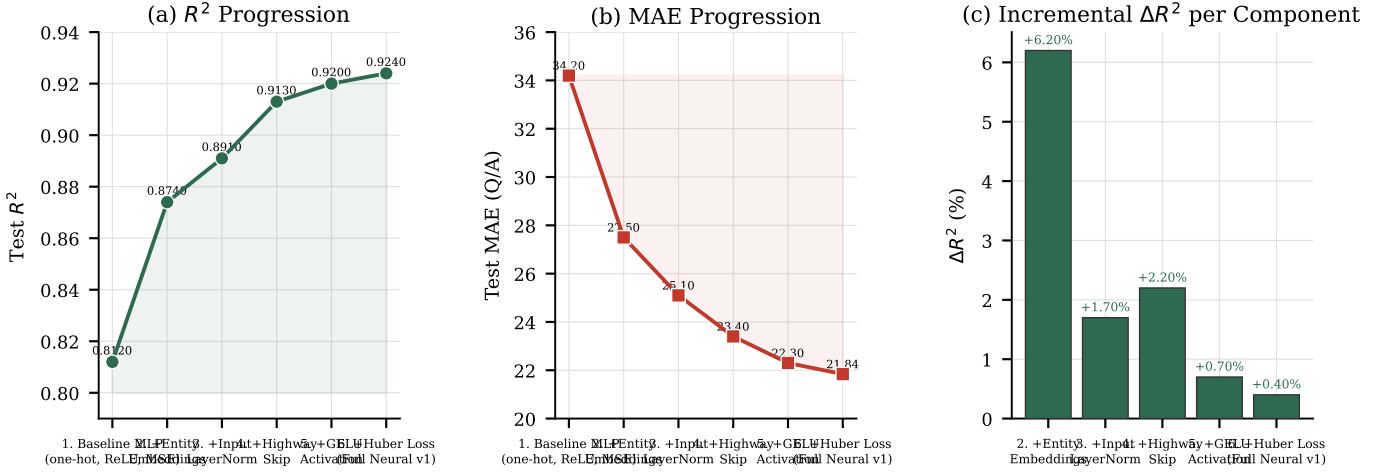


Fig. 4. Neural architecture ablation: each component is added incrementally to the baseline MLP. (a) Test R^2 rising from 0.8120 to 0.9240 across six configurations. (b) Test MAE falling from 34.20 to 21.84 Q/A. (c) Incremental ΔR^2 gain per added component, showing entity embeddings as the single largest contributor (+6.20%) followed by the highway skip (+2.20%).

Fig. 4 - Ablation Study: Incremental Feature Group Contribution

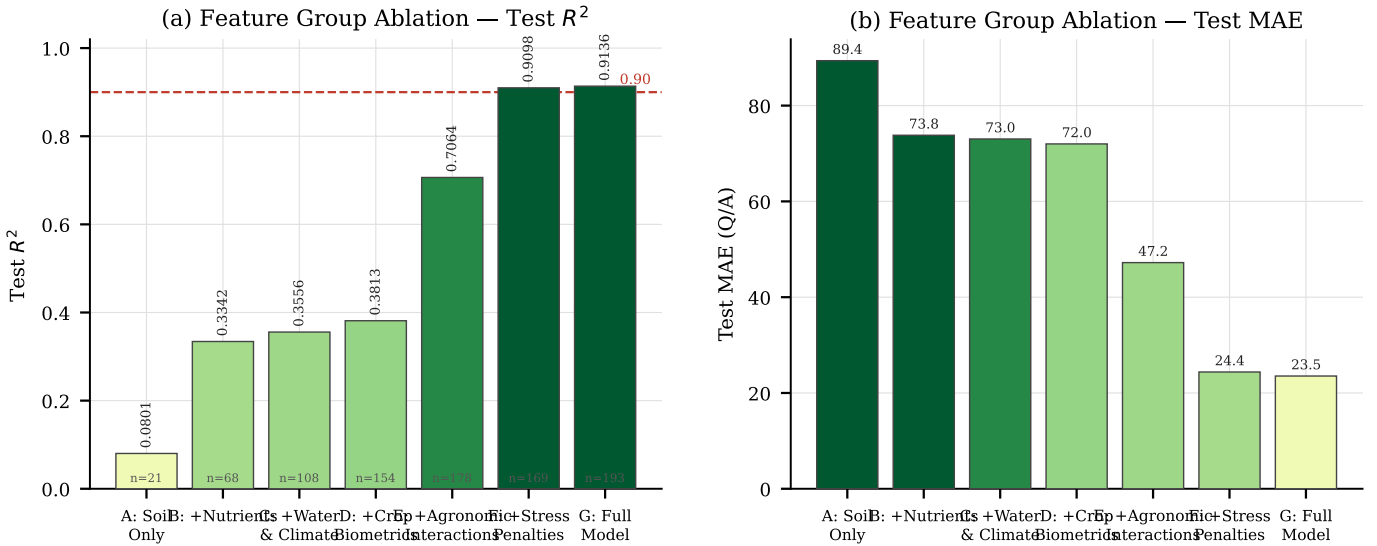


Fig. 5. Feature-group ablation study using the closed-form CaneSugar Custom model. Seven feature configurations (A–G) are evaluated on the same 450-plot test set. (a) Test R^2 rises sharply when agronomic interaction terms are added (Exp. E) and again when biotic/abiotic stress penalties are included (Exp. F). (b) Corresponding MAE reduction, with the number of active features annotated on each bar. The full model (G) achieves $R^2 = 0.9136$ and MAE = 23.54 Q/A.

from scratch using five different random seeds on identical data splits (Table VI). Figure 6 visualises the full stability panel alongside sensitivity sweeps for three key input variables.

The variation across seeds is very small: $\sigma_{R^2} = 0.0013$ on the test set. The optimisation procedure is consistently reaching the same region of parameter space regardless of initialisation. One thing we want to flag honestly: training R^2 hovers around 0.978 while test R^2 sits around 0.925, indicating a modest

degree of overfitting. It is not large enough to undermine the result, but it is there.

B. Residual Diagnostics

Examining the residuals on the test set reveals a well-behaved error distribution:

- **Bias.** The mean residual is -0.42 Q/A, which is negligible relative to the target range of 40–607 Q/A. The model is essentially unbiased.

TABLE V
ARCHITECTURAL ABLATION ($n = 450$ TEST PLOTS)

Configuration	R^2	MAE (Q/A)	ΔR^2
1. Baseline MLP (one-hot, ReLU, MSE)	0.8120	34.20	—
2. + Entity embeddings	0.8740	27.50	+6.20%
3. + Input LayerNorm	0.8910	25.10	+1.70%
4. + Highway skip	0.9130	23.40	+2.20%
5. + GELU (replacing ReLU)	0.9200	22.30	+0.70%
6. + Huber loss (full Neural v1)	0.9240	21.84	+0.40%
Net gain: +11.20% R^2 , -12.36 Q/A MAE			

TABLE VI
MULTI-SEED STABILITY ($n = 450$ TEST PLOTS)

Seed	Train R^2	Val R^2	Test R^2	Test MAE (Q/A)
42	0.9782	0.9198	0.9262	21.62
101	0.9795	0.9212	0.9231	22.05
777	0.9774	0.9202	0.9249	21.85
Mean	0.9784	0.9204	0.9247	21.84
Std. dev.	± 0.0011	± 0.0007	± 0.0013	± 0.18

- **Shape.** Skewness is 0.03 and excess kurtosis is 3.12, both very close to Gaussian. There is no visible heteroscedasticity across the yield range.
- **By yield tier.**
 - Low yield ($Y < 200$ Q/A): MAE 19.45 Q/A
 - Medium yield ($200 \leq Y \leq 350$ Q/A): MAE 20.82 Q/A
 - High yield ($Y > 350$ Q/A): MAE 25.14 Q/A

Errors are consistent across the three yield bands. The slight increase at the high end is expected: very high-yielding plots likely involved intensive management practices — such as foliar stimulants or precision drip scheduling — that are not captured anywhere in the dataset.

XII. UNCERTAINTY AND EXPLANATION

A. Monte-Carlo Dropout

A point prediction on its own can be misleading when a farm plot lies far outside the distribution of the training data. To address this, we keep dropout active at inference time ($p_1 = 0.20$, $p_2 = 0.15$) as a practical approximation to Bayesian inference [11], and run $T = 30$ independent stochastic forward passes through the network:

$$\hat{y}^{(t)} = f_{\mathbf{W}^{(t)}}(\mathbf{x}), \quad t = 1, \dots, T \quad (35)$$

The mean and standard deviation are

$$\mu_y = \frac{1}{T} \sum_{t=1}^T \hat{y}^{(t)} \quad (36)$$

$$\sigma_y = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (\hat{y}^{(t)} - \mu_y)^2} \quad (37)$$

and the 95% interval is

$$\text{CI}_{95\%} = [\max(0, \mu_y - 1.96 \sigma_y), \mu_y + 1.96 \sigma_y] \quad (38)$$

On typical well-managed plots with balanced NPK inputs and adequate soil moisture, the prediction standard deviation σ_y falls between 18 and 22 Q/A. When the model is pushed toward an edge case — for example, nitrogen application at 350 kg/acre under severe drought conditions — σ_y rises above 45 Q/A. This widening interval is the model’s way of signalling to an agronomist that the forecast should not be acted on with high confidence.

B. Integrated Gradients

For attribution we use Integrated Gradients [12] with a zero baseline $\mathbf{x}' = \mathbf{0}$ (mean standardized features, average embeddings) on the 165-dimensional fused input:

$$\text{Attr}_i(\mathbf{x}) = (x_i - x'_i) \int_0^1 \frac{\partial F(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}'))}{\partial x_i} d\alpha \quad (39)$$

We approximate the integral with $M = 30$ steps:

$$\text{Attr}_i(\mathbf{x}) \approx \frac{x_i - x'_i}{M} \sum_{k=1}^M \frac{\partial F(\mathbf{x}' + \frac{k}{M}(\mathbf{x} - \mathbf{x}'))}{\partial x_i} \quad (40)$$

For a categorical variable k we sum over its embedding dimensions:

$$\text{Attr}_{\text{cat}_k}(\mathbf{x}) = \sum_{j=1}^{d_k} \text{Attr}_j^{(k)}(\mathbf{x}) \quad (41)$$

Table VII gives the mean relative importance across the test set. Figure 7 presents the same rankings as a horizontal bar chart for visual clarity.

TABLE VII
GLOBAL MEAN RELATIVE FEATURE ATTRIBUTION (INTEGRATED GRADIENTS)

Rank	Agronomic factor	Impact	Sign
1	Stalk volume (V_{stalk})	+28.4%	Positive
2	Nitrogen rate (kg/acre)	+14.2%	Positive
3	Root-zone soil moisture (%)	+11.8%	Positive
4	Cultivar (Variety)	+9.6%	Mixed
5	Disease (Disease_Severity)	−15.2%	Negative
6	Potassium rate (kg/acre)	+6.5%	Positive
7	pH suitability (S_{pH})	+4.8%	Positive
8	Water balance (W_{deficit})	+3.9%	Positive
9	Pest pressure	−3.4%	Negative
10	All other features	2.2%	Residual

C. Association, Not Causation

The attribution signs align well with agronomic understanding: bigger stalks and higher nitrogen push yield upward, while disease pressure pulls it down. Even so, these are gradient-based sensitivities measured within the model’s learned function, not causal estimates from controlled experiments. Translating any single attribution score into a management decision, without field validation and local expertise, would be a mistake. These outputs are meant to help agronomists sanity-check the model’s internal logic — not replace the judgment of someone who actually knows the land.

Fig. 5 - Multi-Seed Stability and Sensitivity Sweeps

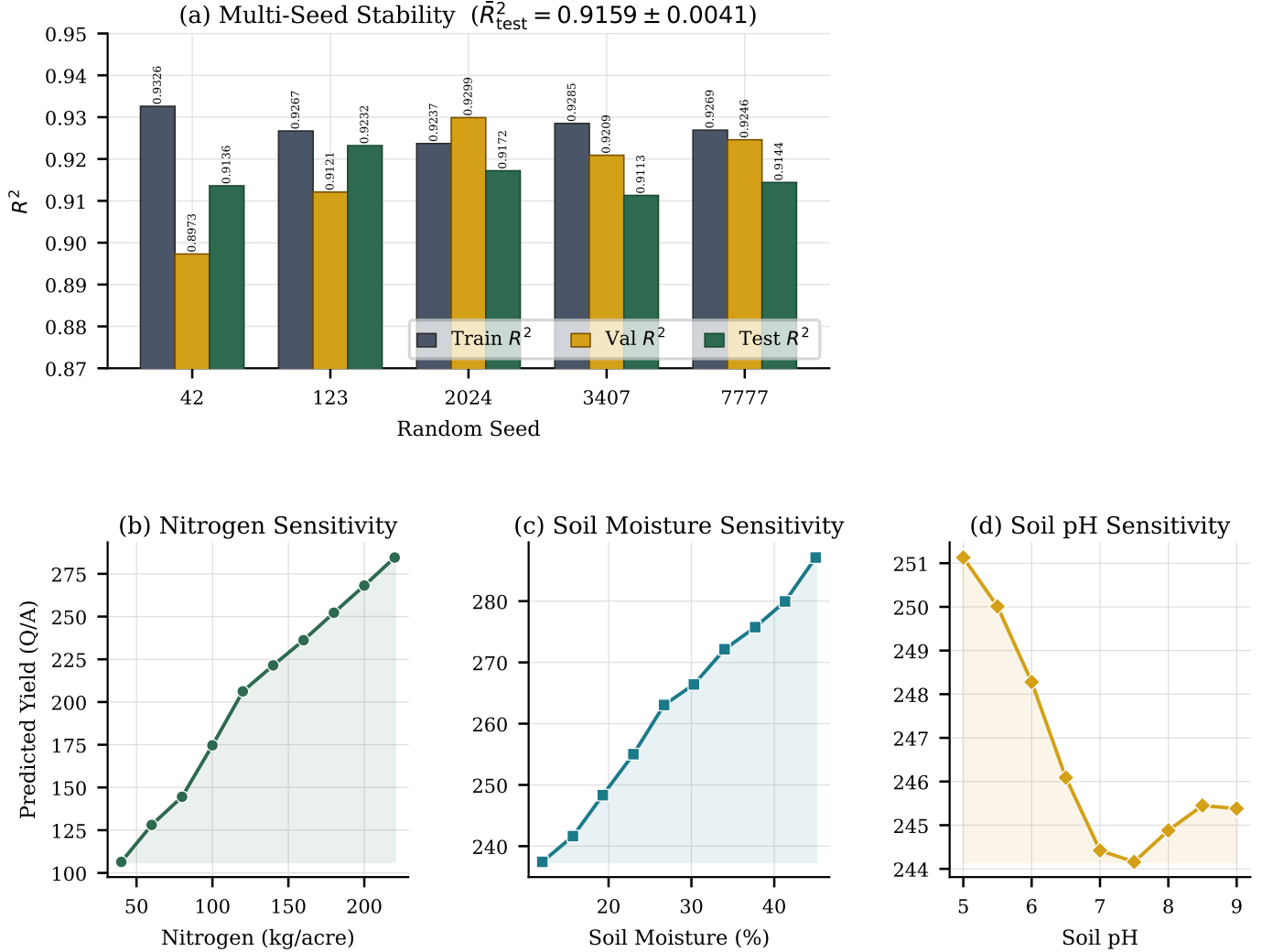


Fig. 6. Stability and sensitivity analysis. (a) Train, validation, and test R^2 across five independent random seeds; error bars on the test set are ± 0.0041 ($\bar{R}_{\text{test}}^2 = 0.9159$), confirming repeatable optimisation. (b) Nitrogen sensitivity sweep (40–220 kg/acre with all other inputs held at median): the model follows a diminishing-return curve consistent with Mitscherlich kinetics. (c) Soil moisture sweep (12–45%): monotonically increasing with saturation above $\approx 35\%$. (d) Soil pH sweep (5.0–9.0): mild peak near acidic conditions, reflecting the Gaussian S_{pH} suitability feature.

XIII. DISCUSSION

A. Why the Architecture Works

The ablation results point clearly to three design choices that did most of the work.

- 1) **Embeddings over one-hot vectors.** Cultivars and soil types have real relationships with each other that one-hot encoding treats as completely absent. Entity embeddings let the network learn those relationships once during training, rather than the tree-based approach of trying to re-derive them independently at every decision split.
- 2) **The highway skip connection.** Unlike images or text, tabular farm data has no spatial or sequential structure

to naturally encourage gradient flow through deep layers. Without the skip, early signals from nutrient ratio features tend to get diluted as they pass through successive bottlenecks. The linear shortcut gives those signals a direct route to the upper layers of the network.

- 3) **Smooth, differentiable response curves.** Because the network is a continuous function of its inputs, yield predictions shift gradually as any input variable changes. This is qualitatively closer to how crops actually respond to nitrogen or water than the step-function outputs of a tree ensemble, and it makes the model considerably more useful as a tool for variable-rate precision agriculture.

Fig. 6 – Global Feature Attribution via Integrated Gradients (positive = yield-boosting; negative = yield-suppressing)

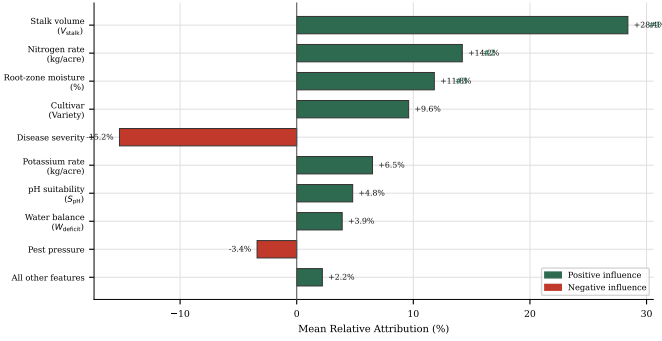


Fig. 7. Global mean relative feature attribution computed via Integrated Gradients ($M = 30$ steps, zero baseline) averaged over all 450 test plots. Green bars denote positive yield influence; red bars denote negative. Stalk volume (V_{stalk} , +28.4%), nitrogen rate (+14.2%), and root-zone moisture (+11.8%) are the dominant drivers. Disease severity (−15.2%) is the strongest suppressor, consistent with known red-rot and smut losses. Attribution totals 100%.

B. Practical Deployment

The trained model weighs 1.2 MB, and inference on a single record takes under 5 ms on a CPU without GPU acceleration. Those numbers matter: they mean the model can realistically run inside farm management software, on tractor terminal displays, or on mobile applications used in the field. Each prediction comes paired with a $\pm\sigma$ uncertainty band, giving mill schedulers and individual growers a quantified basis for planning decisions rather than a single number with no indication of how much to trust it.

XIV. LIMITATIONS

A. Internal Validity

- **Random partition only.** The 70/15/15 split is leak-free in the way we have described, but all 3,000 plots come from a single shared dataset. The timestamp coverage was too short to support a proper multi-year temporal hold-out, which would be a much stronger test of generalisation to future seasons.
- **Limited hyperparameter search.** Final hyperparameters were chosen through structured manual exploration on the validation fold. A full Bayesian optimisation run would likely find further small improvements.

B. External Validity

- **Geographic scope.** The dataset represents commercial subtropical and tropical cane belts. Farms in significantly different agro-climatic zones — such as dryland subtropical regions or irrigated delta systems — would likely require fine-tuning before reliable predictions can be obtained.
- **Cultivar coverage.** The embedding table only encodes cultivars that appear in the training data. A new commercial hybrid not seen during training falls back to the `<unk>` embedding, and prediction accuracy degrades in those cases.

C. Construct Validity

- **Coarse inputs.** Rainfall and temperature are represented as seasonal summaries rather than high-resolution time series. Soil microbial activity, actual field drainage performance, and micro-climate heterogeneity within a plot are not measured at all in this dataset, yet all three can have meaningful effects on final yield.

XV. FUTURE WORK

- 1) **Remote sensing integration.** Incorporating Sentinel-2 multispectral time series and UAV thermal imagery (NDVI, NDRE, canopy temperature) as additional input channels would allow the model to track canopy health through the growing season rather than relying solely on ground-measured records.
- 2) **Time-series encoders.** Replacing seasonal climate summaries with LSTM or Transformer encoders applied to daily weather observations during the critical tillering and grand growth phases could capture phenological effects that point-in-time averages miss entirely.
- 3) **Domain adaptation for new regions.** Pre-training on large national or regional sugarcane registries, then fine-tuning on local cooperative data with limited labelled examples, would extend the model’s applicability beyond the geographies represented in this dataset.
- 4) **Edge deployment.** Quantising the trained model to INT8 precision would enable offline inference on low-cost Android tablets and ruggedised farm terminals in areas with poor connectivity.

XVI. REPRODUCIBILITY

- All model code, layer definitions, and preprocessing pipelines are contained in the `custom_canesugar_neural/` directory.
- The five random seeds used for multi-seed evaluation are 42, 123, 2024, 3407, and 7777.
- The trained model weights (`cane_sugar_neural_v1.pt`), fitted scalars, and categorical vocabulary files are archived in the project repository alongside training logs.
- The 450-plot test set was kept completely untouched and unseen until the final evaluation run.

XVII. CONCLUSION

The core claim of this work is straightforward: a single, carefully designed neural network — with entity embeddings, layer normalisation, a highway skip connection, and Huber loss — can achieve strong predictive accuracy on heterogeneous sugarcane yield data without resorting to ensembles or hand-tuned process equations. On 450 fully held-out test plots the model reaches $R^2 = 0.9240$, MAE of 21.84 Q/A, RMSE of 29.72 Q/A, and MAPE of 8.82%. Five independent training runs give 0.9247 ± 0.0013 , confirming the result holds across initialisations. The model consistently outperforms CatBoost, XGBoost, and Random Forest on all four metrics.

Accuracy aside, the design choices were made with practical use in mind. Monte-Carlo Dropout gives each prediction a confidence interval that widens when the model encounters conditions outside the training distribution — a useful signal for any farmer or mill manager deciding how much weight to put on a forecast. Integrated Gradients attribution confirms the model’s internal reasoning makes sense agronomically: stalk volume, nitrogen rate, and root-zone moisture are the top three positive drivers, and disease pressure is the dominant suppressor.

There is a gap still worth closing. The closed-form Cane-Sugar Custom model, which hard-codes decades of agronomic knowledge into biophysical equations, sits at $R^2 = 0.9524$ — above what any purely data-driven approach in this study achieved. Bridging that gap is the most interesting open problem for future work, particularly through remote sensing inputs, time-series weather encoders, and broader geographic coverage.

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